



Figure 3: R icon after installation



Figure 4: RStudio icon

date and logical. The following code shows how to use various data types:

```
> x<-"data"
> class(x)
[1] "character"
> nchar(x)
[1] 4
> d<-as.Date("2017-12-01")
> d
[1] "2017-12-01"
> class(d)
[1] "Date"
> b<-TRUE
> class(b)
[1] "logical"
```

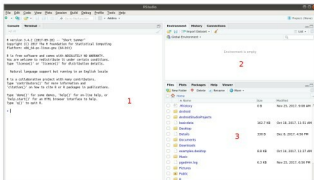


Figure 5: RStudio default screen

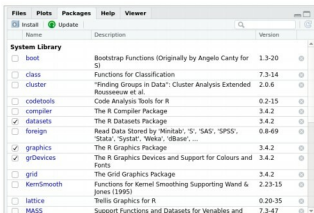


Figure 6: Packages in R Studio

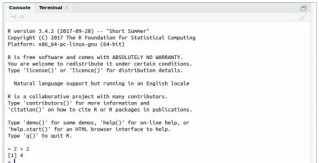


Figure 7: Using the console in R Studio

Apart from basic data types, R supports data structures or objects like vectors, lists, arrays, matrices and data frames. These are the key objects or data structures in R.

A vector stores data of the same type. It can be thought of as a standard array in most of the programming languages. A 'c' function is used to create a vector ('c' stands for 'combine').

The following code snippet shows the creation of a vector:

```
> v <- c(10,20,30,40)
> v
[1] 10 20 30 40
```

The most interesting thing about a vector is that any operation applied on it will be applied to individual elements of it. For example, 'v + 10' will increase the value of each element of a vector by 10.

```
> v + 10
[1] 20 30 40 50
```

This concept is difficult to digest for some, but it's a very powerful concept in R. Vector has no dimensions; it is simply a vector and is not to be confused with vectors in mathematics which have dimensions. Vector can also be created by using the ':' sign with start and end values; for example, to create a vector with values 1 to 10, use 1:10.

```
> a <- 1:10
> a
[1] 1 2 3 4 5 6 7 8 9 10
```

It is also possible to do some basic operations on vectors, but do remember that any operation applied on a vector is applied on individual elements of it. For example, if the addition operation is applied on two vectors, the individual elements of the vectors will be added:

```
> a<-1:5
> b<-21:25
> a+b
[1] 22 24 26 28 30
```